1 **(a)** **(a)** $8000(1+\frac{p}{100})^3 = 27000$ **[M1]**

$1+\frac{p}{100} = \sqrt[3]{\frac{27000}{8000}} = \frac{3}{2}$ **[M1]**

50 **[A1]**

(b) $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Total Marks: 5]

2 $450 \times \left(1+\frac{6.5}{100}\right)^{12}$ **[M1]**

450×1.065^{12} **[M1]**

958.01 (accept 958) **[A1]**

[Total Marks: 3]

3 $32000 \times (1-0.12)^5$ oe **[M2]**

For 32000×0.88^n where $n > 1$, award **[M1]**

16887 **[A1]**

[Total Marks: 3]

4 **(a)** $65000 \times \left(1+\frac{2.8}{100}\right)^5$ or 65000×1.028^5 **[M1]**

74624 **[A1]**

(b) $\frac{74624-65000}{65000} \times 100$ or $(1.028^5 - 1) \times 100$ **[M1]**

14.8 **[A1]**

[Total Marks: 4]

5 **(a)** **(a)** $85000(1+\frac{r}{100})^{12} = 112000$ **[M1]**

$1+\frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

(b) $85000 \times \left(1+\frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1+\frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Total Marks: 5]

6 **(a)** $45000 \times \left(1 - \frac{x}{100}\right)^4 = 26000$ **[M1]**

$1 - \frac{x}{100} = \sqrt[4]{\frac{26000}{45000}}$ or $1 - \frac{x}{100} = 0.8711...$ **[M1]**

12.9 **[A1]**

(b) $45000 \times (0.8711...)^n < 10000$ or $(0.8711...)^n < \frac{2}{9}$ **[M1]**

Testing values or using logarithms: $n = 10 \implies \$11342$ and $n = 11 \implies \$9881$ **[M1]**

11 **[A1]**

[Total Marks: 6]

7 **(a)** $28\,500 \times \left(1 - \frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

(b) $35\,000 \times \left(1 - \frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1 - \frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1 - \frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Total Marks: 7]

8 **(a)** Let $k = 1 - \frac{r}{100}$. Then $M_0k^2 = 80$ and $M_0k^5 = 40.96$ **[M1]**

$k^3 = \frac{40.96}{80} = 0.512 \implies k = \sqrt[3]{0.512} = 0.8$ **[M1]**

$1 - \frac{r}{100} = 0.8 \implies r = 20$ **[A1]**

(b) $M_0 \times 0.8^2 = 80$ or $M_0 \times 0.64 = 80$ **[M1]**

125 **[A1]**

(c) $125 \times 0.8^n < 10$ or $0.8^n < 0.08$ **[M1]**

Testing values: $n = 11 \implies 0.8^{11} = 0.0858...$ and $n = 12 \implies 0.8^{12} = 0.0687...$ **[M1]**

12 **[A1]**

[Total Marks: 8]

9 **(a)** $2500 \times \left(1 + \frac{r}{100}\right)^6 = 9100$ **[M1]**

$\left(1 + \frac{r}{100}\right)^6 = 3.64$ **[M1]**

$1 + \frac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept $24.03\ldots$) **[A1]**

(b) 2500×1.2403^{10} or 9100×1.2403^{4} **[M1]**

21 500 **[A1]**

(c) $2500 \times 1.2403^{n} > 50\,000 \implies 1.2403^{n} > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

[Total Marks: 9]

10 $26000 \times (1 - 0.15)^{4}$ or 26000×0.85^{4} **[M2]**

If M0, allow [M1] for 26000×0.85^{k} where $k > 1$

13572 **[A1]**

[Total Marks: 3]

11 (a) 1500×1.08^{12} **[M1]**

3777 **[A1]**

(b) $1500 \times 1.08^{n} > 3000$ or $1.08^{n} = 2$ **[M1]**

Trial with values (e.g., $1.08^{9} = 1.999...$ and $1.08^{10} = 2.158...$) or uses logarithms (e.g., $n = \frac{\log(2)}{\log(1.08)} = 9.006...$) **[M1]**

10 **[A1]**

[Total Marks: 5]

12 (a) $\left(1 + \frac{r}{100}\right)^{2} = 1.44$ or $x^{2} = 1.44$ **[M1]**

$1 + \frac{r}{100} = 1.2$ or $x = 1.2$ **[M1]**

20 **[A1]**

(b) Number of doubling periods $= 45 \div 15 = 3$ or 80×2^{3} **[M1]**

640 **[A1]**

[Total Marks: 5]